PERFORMANCE ANALYSIS OF THE SBAS ALGORITHM FOR SURFACE DEFORMATION RETRIEVAL

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1 ABSTRACT

We investigate the performance of the Differential Synthetic Aperture Radar Interferometry (DInSAR) technique referred to as Small BAseline Subset (SBAS) [1] approach, which allows the generation of mean deformation velocity maps and displacement time series. In particular, we have analyzed the SBAS procedure performance by processing ERS SAR data acquired by the European Remote Sensing Satellite (ERS) sensors and comparing the achieved results with geometric leveling and GPS measurements. The analysis has been focused on the Napoli bay (Italy) and Los Angeles (California) test areas where different deformation phenomena are present and a large amount of ERS SAR data and geometric leveling (in the Napoli zone) and continuous GPS (in the Los Angeles zone) measurements are available. The carried out analysis shows that the SBAS technique allows to achieve an estimate of the mean deformation velocity values with a standard deviation of about 1mm/year and, for the single displacement measurements, with a standard deviation of about 5mm.

2 INTRODUCTION

Differential Synthetic Aperture Radar Interferometry (DInSAR) is a well-established remote sensing technique [2] that allows to investigate surface deformation phenomena by exploiting the phase difference (usually referred to as interferogram) of two SAR images relevant to temporally separated observations of an investigated area. An effective way to detect and follow the temporal evolution of deformations is via the generation of time series; in this case, the information available from each interferometric data pair must be properly related to those included in the other acquisitions by generating an appropriate sequence of DInSAR interferograms.

In this context, several approaches aimed at the DInSAR time series generation have been proposed. In this work, we focus on the technique referred to as Small BAseline Subset (SBAS) algorithm [1], that has been originally developed to investigate large spatial scale displacements with relatively low resolution (typically of the order of 100 x 100 m). The SBAS approach relies on the use of small baseline differential SAR interferograms and on the application of the singular value decomposition (SVD) method in order to link independent SAR acquisition data set, separated by large baselines, thus increasing the number of data used for the analysis. The capability of this approach to generate deformation maps and time series from data acquired by the European Remote Sensing Satellite (ERS) sensors have been already shown in different applications [3, 4, 5]; however, no exhaustive analysis on the quality of the DInSAR measurements has been yet provided. Accordingly, we investigate in this work the SBAS algorithm performance by carrying out an extensive SAR/geodetic measurements comparison on two test areas where different deformation phenomena are present and, at the same time, a large amount of SAR data are available as well as ground measurements.

In particular, the first test site is located in the Napoli bay (Italy) area which includes three active volcanoes (the Campi Flegrei caldera, the Somma-Vesuvio volcanic complex and the Ischia island) and the highly urbanized city of Napoli. In this area a large number of ERS-1/2 SAR data, acquired in the 1992-2003 time period and relevant to ascending and descending orbits, was available to us. The availability of these two SAR data set permitted to analyze the temporal behavior of the detected deformation projections along the two radar line of sight (LOS) and, at the same time, to discriminate east-west and vertical displacement components. Moreover, the geodetic measurements obtained through the geometric spirit leveling network present in the region, selected within the same time interval of the SAR acquisitions, allowed us to perform an extensive comparison between the DInSAR and the leveling measurements.

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The second test site is the Los Angeles (California) metropolitan zone, characterized by surface deformations that are a combination of natural and anthropogenic signals. Also in this case a large amount of ERS images acquired on descending orbits, and of geodetic measurements recorded through the Southern California Integrated GPS Network (SCIGN), is available. Accordingly, a detailed comparison between the DInSAR and the GPS measurements was carried out.

3 DInSAR AND SPIRIT LEVELING MEASUREMENTS COMPARISON: THE NAPOLI BAY CASE STUDY

The results presented in the following section are relevant to the comparison between the retrieved SBAS-DInSAR deformation measurements and those available from the geometric leveling campaigns. In particular, our analysis is focused on the portion of the Napoli bay area extending from the NW border of the Campi Flegrei caldera up to the SE margin of the Somma-Vesuvio volcanic complex, thus including the entire Napoli city area, see Fig.1a.
In this zone one of the most extended geometric leveling network of the world, with about 600 benchmarks and more than 350 km of linear extension, is present [6, 7, 8]. This network, managed by the Osservatorio Vesuviano (OV) belonging to the Italian National Institute of Geophysics and Volcanology (INGV), is a key element of the surveillance system of this area characterized by a high risk degree. In Fig. 1a we present the geocoded SAR amplitude image of the investigated zone with superimposed the locations of all the benchmarks (squares) of the existing leveling network and with highlighted (colored squares organized in six different lines) the part of the network for which geodetic measurements were made available to us. All the used leveling data span the 1990-2003 time interval (because of their temporal overlap with the overall available SAR data set) and have been referred to the same benchmark located in the Napoli city, marked by LNA001 in Fig. 1b.

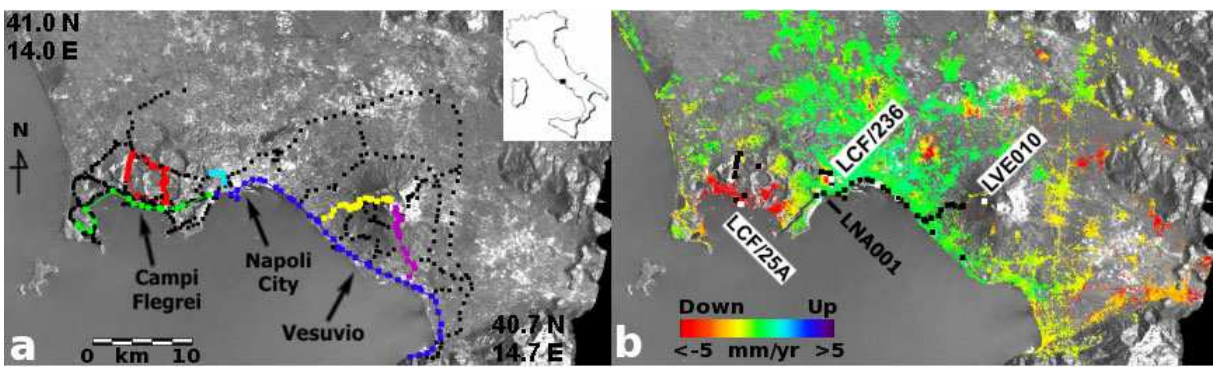

Fig. 1. Investigated portion of the Napoli bay (Italy) area. a) SAR amplitude image of the zone with highlighted the main locations. The squares mark the benchmarks positions of the existing leveling network; the lines identified by different colored squares represent a selection of the leveling benchmarks for which the geodetic measurements were available to us. b) Vertical mean deformation velocity map with superimposed the locations of the leveling benchmarks (black and white squares) characterized by a dominantly vertical displacement. White squares mark the selected benchmarks relevant to the plots shown in Figs. 2g-i; the benchmark LNA001 identifies the reference point for both DInSAR and geodetic measurements.

Concerning the SAR data analysis, we used in our study 116 SAR data acquired by the ERS-1/2 sensors. In particular, 58 acquisitions are relevant to ascending passes (track: 129, frame: 809) spanning the 1993 - 2003 time interval; the remaining 58 data have been acquired from descending orbits (track: 36, frame: 2781) between the 1992 and the 2002. Starting from the available SAR data set, we have computed 133 and 148 interferograms from the acquisitions relevant to the ascending and descending orbits, respectively. Each interferometric SAR image pair has been chosen with a perpendicular baseline value smaller than 300 m and with a maximum time interval of 4 years; precise satellite orbital information and a Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM) [9] of the area have also been used. All the DInSAR products have been obtained following a complex multilook operation [10] with 20 looks in the azimuth direction and 4 looks in the range one, with a resulting pixel dimension of the order of 100 x 100 m.
As a first result achieved by applying the SBAS approach, we generated the LOS mean deformation velocities relevant to the ascending and descending orbits. Moreover, by benefiting of the availability of these two maps, we have discriminated the vertical and east-west displacement components similarly to what done in [4, 5]. In particular, the

difference between the mean surface velocity maps computed from the ascending and descending orbits on pixels common to both maps, allowed us to get an estimate of the surface deformation velocities in the east-west direction; the sum of the ascending and descending deformation patterns provided information on the vertical motion. In Fig. 1b, we show the retrieved vertical mean deformation velocity, geocoded and superimposed on the SAR amplitude image of the zone. This map has been computed with respect to a reference pixel located in correspondence of the leveling benchmark LNA001; note also that noisy areas (non coherent zones) with low accuracy measurements have been excluded from this analysis.

The availability of the estimated vertical deformation component shown in Fig. 1b represents a key element of our study because it allows us to compare the obtained DInSAR results with the vertical displacement measurements available from the leveling network in correspondence of the pixels located in coherent areas common to both ascending and descending DInSAR maps. In particular, we have carried out the following two operations: we first compared the mean vertical displacement velocities computed from the SAR and the geodetic data; then, we compared the temporal evolution of the deformations detected from both the ascending and descending orbits, in areas characterized by a nearly vertical motion, with those computed from the leveling measurements.

As concerns the first issue of the SBAS performance analysis, our comparison was focused on the benchmarks of the leveling network identified by the six colored lines shown in Fig. 1a. The achieved results are presented in the plots of Figs. 2a-f and they clearly show the good agreement between the two vertical mean velocity measurements (SAR: red triangles, leveling: black stars). Moreover, for all the coherent pixels associated to the leveling benchmarks, we have computed the standard deviation value, say σ_v, of the difference between SAR and leveling vertical velocities (the latter computed within the temporal interval common to the SAR data), obtaining $\sigma_v = 1.0$ mm/year, corresponding in LOS to about 0.9 mm/year.

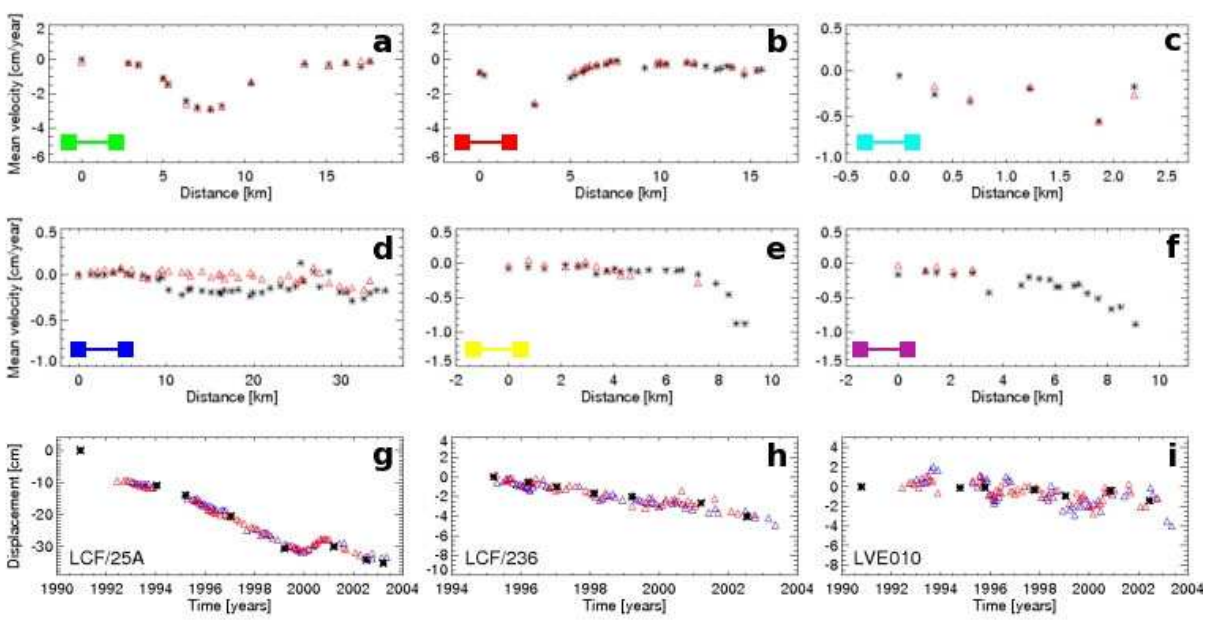

Fig. 2. SAR/leveling comparisons. a-f) Plots of the vertical mean deformation velocities computed from the SAR data (red triangles) and compared to the corresponding ones relevant to the leveling measurements (black stars) along the lines identified in Fig. 1a by the green, red, cyan, blue, yellow and violet squares, respectively. g-i) Comparison between the DInSAR LOS deformation time series (ERS ascending data: blue triangles, ERS descending data: red triangles) and the corresponding leveling measurements projected on the radar LOS (black stars) for the pixels labeled in Fig. 1b as LCF/25A, LCF/236 and LVE010, respectively.

Regarding the second issue of our comparison, we have identified all the leveling benchmarks characterized by an horizontal mean deformation velocity significantly small (in particular, with an amplitude smaller than 2.5 mm/year) and located in correspondence of coherent areas in both ascending and descending maps. The leveling pixels identified following this operation are those marked by the black and the white squares superimposed to the vertical mean

deformation velocity shown Fig. 1b. For these pixels we have compared the deformation time series computed from both the ascending and descending SAR acquisitions with the leveling measurements projected on the radar line of sight (a 23° mean incidence angle was assumed). In Figs. 2g-i, we show the plots relevant to the DInSAR/leveling comparison for a selection involving the three sample benchmarks identified in Fig. 1b by the white squares labeled as LCF/25A, LCF/236 and LVE010, respectively. The strong similarity between the temporal evolution of the ascending (blue triangles) and descending (red triangles) SAR measurements, as well as their agreement with the LOS-projected leveling (black stars) deformation time series is evident. Moreover, we have computed, and subsequently averaged, for all the benchmarks identified in Fig. 1b by the black and white squares, the standard deviation values, say σ_{d_L}, of the differences between the SAR and the LOS-projected leveling measurements, the latter interpolated via a linear regression within the interval common to the SAR data. Following this analysis, we obtained $\sigma_{d_{L_A}} = 4.6$ mm, for the ascending, and $\sigma_{d_{L_D}} = 4.8$ mm, for the descending data. These two values are consistent and allows us to conclude that the average value $\sigma_{d_L} = 4.7$ mm represents a realistic estimate for the standard deviation of the difference between SAR and leveling data.

As a final remark, we stress that in our analysis we have assumed the leveling data as a reference, without considering the errors that can affect these measurements. If we reconsider this issue, we may remove the bias due to the estimated errors relevant to the leveling measurements, by referring to the typical accuracy of the geometric leveling data that in this study is assumed equal to 2 mm [9]. Following this bias removal, we finally obtain the standard deviation value $\sigma_{d_L} = 4.3$ mm for the difference between SAR and leveling data.

4 DInSAR AND GPS MEASUREMENTS COMPARISON: THE LOS ANGELES METROPOLITAN AREA CASE STUDY

The following section is dedicated to present the results of the comparison between the SBAS-DInSAR results and the measurements available from continuous GPS measurements. In particular, the investigated test site is represented by the Los Angeles metropolitan zone (Southern California, USA), that is a tectonically active region with surface deformations caused by a variety of natural and anthropogenic signals. We remark that a key element for the seismic surveillance on the whole area is represented by the Southern California Integrated GPS Network [11] which is an array of 250 GPS stations spread out across southern California and northern Baja California, Mexico.

For what concerns the presented DInSAR analysis, the SBAS algorithm has been applied to a set of 42 SAR data (track: 170, frame: 2925), acquired by the ERS satellites in the 1995-2002 time interval and coupled in 102 interferograms. The processing parameters used in this case are consistent with those applied for the generation of the DInSAR products relevant to the previous analysis of the Napoli bay site (see section 3). Note also that in this case we have reconsidered the results of the study presented in [4], that was originally focused on the Santa Ana basin aquifer.

As a first result of the SBAS algorithm analysis, we present in Fig. 3a the geocoded SAR amplitude image relevant to the investigated area, with superimposed the retrieved LOS mean displacement velocity map and all the locations of the GPS SCIGN sites (black and white squares) within the region. Note also that, in our analysis we selected the GPS stations located in coherent areas, and for which the measurements are relevant to a time interval starting before 2000 (in order to ensure at least two years of overlap with the available SAR data).

Following the mentioned GPS selection, we have compared the DInSAR time series with the corresponding LOS-projected GPS measurements, the latter obtained through the SCIGN Web-Site [12]. In Figs. 3b-g, we show the plots relevant to the DInSAR/GPS comparison for six sample GPS stations identified in Fig. 3a by the white squares and labeled as CIT1, CVHS, WHC1, LBC2, SACY and FVPK, respectively. Note that both DInSAR and geodetic measurements have been referred to the same pixel located in correspondence of the ELSC station (see Fig. 3a). The presented results clearly show the good agreement between these two measurements.

Similarly to what done in section 3, we have computed and then averaged the standard deviation values of the differences between the DInSAR and the LOS-projected GPS time series relevant to all the GPS sites identified in Fig. 3a; in this case, we obtained $\sigma_{d_G} = 6.9$ mm. Subsequently, we computed the LOS-projected GPS errors available from the SCIGN Web-Site [12] and removed the bias due to the estimated errors relevant to the geodetic measurements. Following this bias removal, we finally achieved the value $\sigma_{d_G} = 5.6$ mm for the standard deviation of the difference between DInSAR and GPS data. This result is consistent with what previously found in section 3 for the DInSAR/leveling comparison.

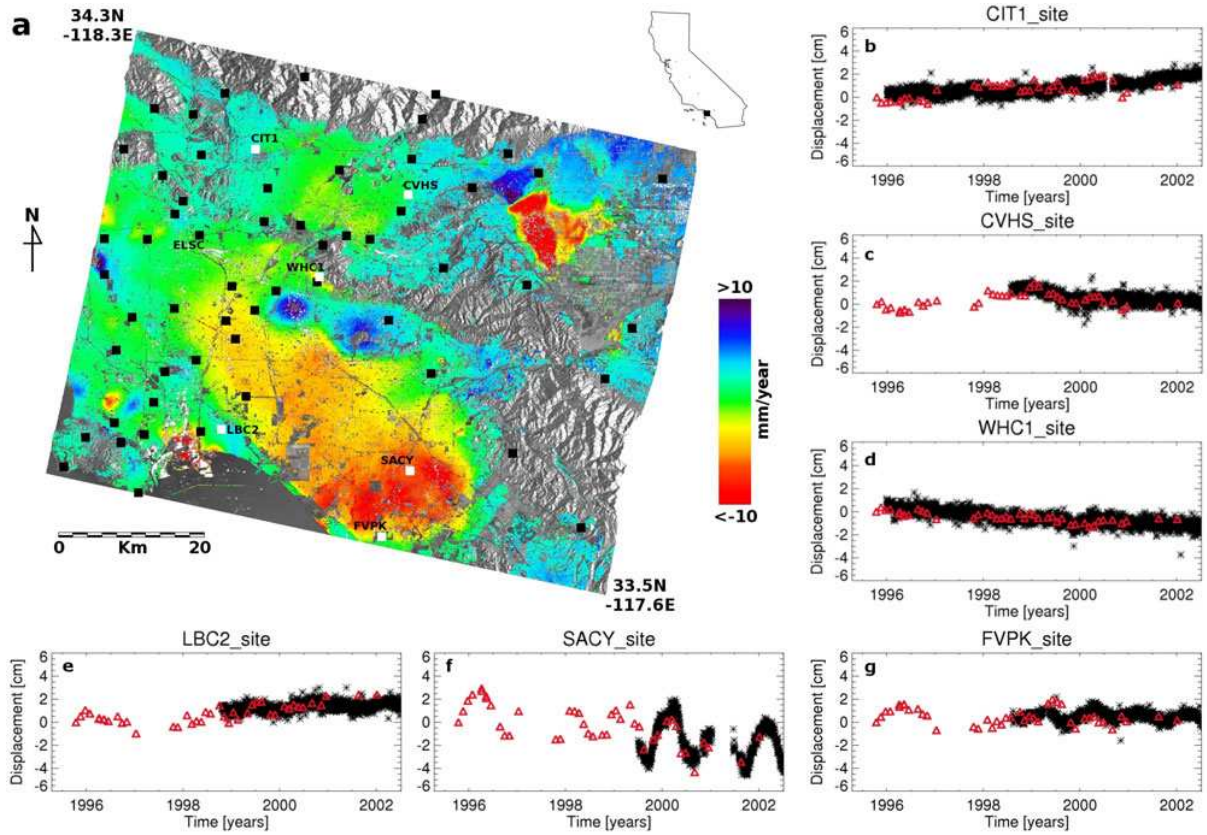

Fig. 3. DInSAR results relevant to Los Angeles (California) metropolitan area. a) LOS mean deformation velocity map with superimposed the locations of the investigated GPS stations (black and white squares). White squares mark the selected stations relevant to the plots shown in Figs. 3b-g; the label ELSC identifies the reference point for both DInSAR and geodetic measurements. b-g) Comparison between the DInSAR LOS deformation time series (red triangles) and the corresponding GPS measurements projected on the radar LOS (black stars), for the pixels labeled in Fig. 3a as CIT1, CVHS, WHC1, LBC2, SACY and FVPK, respectively.

5 CONCLUSIONS

The presented study demonstrates the capability of the SBAS algorithm to provide an estimate of the mean deformation velocity value with a standard deviation of about 1mm/year. Moreover, we have also shown that a value of $\sigma_d = 5$ mm represents a very realistic number for what concerns the standard deviation of the single measurements of the retrieved DInSAR LOS deformation time series.

As a final remark, we stress that a more detailed analysis on the SBAS algorithm performance can be found in [13]. In this work it is also presented a first result showing that there is a dependence of the error estimates with respect to the distance from the reference pixel with an increase of the standard deviation values with a spatial variation of about 0.05 mm/km. This is an additional issue to be taken properly into account when considering the accuracy of the DInSAR deformation measurements.

6 ACKNOWLEDGMENTS

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(http://www.scign.org/). Moreover, the DEMs of the investigated zones have been achieved through the SRTM archive while precise ERS-1/2 satellite orbit state vectors are courtesy of the Technical University of Delft, The Netherlands. The authors also want to thank the Osservatorio Vesuviano which has provided the leveling measurements of the Napoli bay site and in particular dr. G. P. Ricciardi for his continuous help and support.

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